

The Ecological Value and Spatial Reconstruction of Integrating Green Energy into Modern Commercial Space Design

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Abstract: As global sustainable development advances, green energy plays an increasingly important role in modern commercial space design. This paper discusses the ecological value and spatial transformation brought by applying green energy to such design, and analyzes its effects on ecological benefits as well as the restructuring of spatial functions and forms. It first introduces the concept of green energy, explains its importance in modern architecture and its inherent link with commercial space planning. Combined with theoretical analysis and case studies, the paper further explores the ecological strengths of green energy in commercial spaces, such as higher energy efficiency, lower carbon emissions and better ecological service performance.

Keywords: Green energy; Commercial space design; Ecological value; Space reconstruction; Sustainable development

1. Introduction

1.1 Research Significance

Integrating green energy into the design of modern commercial spaces contributes to sustainable development and has a positive impact on the ecological environment. As global warming and resource scarcity become increasingly severe, the traditional energy-intensive and inefficient design models for commercial spaces can no longer meet modern society's requirements for environmental protection and energy conservation [1].

By utilizing renewable resources such as solar and wind energy, commercial spaces can effectively reduce carbon emissions while also playing a positive role in ecosystem conservation. For example, the application of green energy technologies can optimize building energy consumption patterns, decrease reliance on fossil fuels, and thereby mitigate urban heat island effects and environmental pollution.

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The integration of green energy in spatial functional restructuring transforms commercial spaces from single-purpose consumption venues into multi-functional complexes. Traditional commercial space design has predominantly focused on economic benefits while neglecting the balance between ecological advantages and social value [2]. The adoption of green energy introduces innovative features in spatial layout, functional zoning, and user experience, utilizing natural lighting, ventilation systems, and green roofs to enhance environmental quality while fostering greater human-nature interaction. Such spatial reconfiguration not only improves commercial space utilization efficiency but also encourages consumer acceptance and participation in sustainable lifestyles, thereby fostering a better human-environment relationship. Furthermore, green energy applications open up new innovation opportunities for commercial spaces—such as intelligent energy management systems and distributed energy networks—thereby expanding their functional boundaries.

1.2 Research Methods and Technical Approach

This study employs a combined research methodology integrating literature analysis, case studies, and spatial simulation techniques to ensure systematic rigor and scientific validity. First, literature analysis was conducted to review advancements in green energy and commercial space design, establishing the theoretical foundations and practical context of the field. The analysis primarily referenced research on ecological value reconstruction and spatial optimization [3], from which the study developed an ecological value-reconstruction-based framework for rural settlement spatial optimization, providing crucial methodological insights. Furthermore, by examining current trends in green energy applications within commercial spaces, the research expanded its scope and laid a theoretical foundation for future investigations. Through case studies of representative commercial projects, the study identified key elements and implementation approaches for integrating green energy into design, thereby validating the feasibility and effectiveness of the proposed theoretical model.

2. Theoretical Foundations of Green Energy and Commercial Space Design

2.1 The Concept and Development History of Green Energy

Green energy refers to renewable energy sources that have minimal environmental impact during production and use, including solar, wind, hydro, and biomass energy. These sources

are characterized by their cleanliness, sustainability, and low emissions, making them an effective approach to addressing global climate change and energy crises [4]. With technological advancements and policy support, the application of green energy has expanded worldwide, serving as a crucial driver for economic transformation and sustainable development. In commercial space design, adopting green energy not only reduces energy consumption but also enhances the ecological value and user experience of the space. For instance, the use of solar photovoltaic panels and natural lighting significantly decreases reliance on traditional energy sources while creating more comfortable and environmentally friendly environments for occupants.

The development of green energy has evolved from initial experiments to widespread adoption. Early efforts primarily focused on hydropower and wind energy, but in recent years, with technological advancements and cost reductions, renewable energy sources such as solar power have gradually become mainstream. Research indicates that the adoption of green energy is influenced not only by technical factors but also by policy support, market mechanisms, and public awareness [5]. In commercial space design, the application of green energy should be tailored to specific contexts to achieve optimal ecological and economic benefits.

2.2 Core Concepts of Commercial Space Design

Modern commercial space design centers on functionality, environmental sustainability, and user experience, emphasizing the integration of spatial layout with its surroundings. With societal development and evolving consumer preferences, commercial spaces have evolved from mere shopping venues into comprehensive

complexes that combine leisure, social interaction, and cultural experiences. The design philosophy adopted not only prioritizes spatial functionality and aesthetics but also underscores the importance of human-nature interaction, striving for efficient, flexible, and eco-friendly layouts. The use of green energy serves as a crucial means to enhance the sustainability of commercial spaces, thereby opening new directions and opportunities for design

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3. The Multidimensional Manifestation of Ecological Value

3.1 Improvement of Energy Efficiency and Reduction of Carbon Emissions

The application of green energy in commercial spaces serves as an effective approach to optimize energy efficiency and achieve low-carbon operations. By integrating renewable energy systems such as solar and wind power, commercial spaces can significantly reduce their reliance on traditional fossil fuels and enhance overall energy utilization. Building-Integrated Photovoltaics (BIPV) technology embeds solar panels into building exteriors or roofs, minimizing energy loss during transmission while enabling localized production and consumption. Additionally, intelligent energy management systems monitor equipment status in real time and adjust operational parameters as needed, ensuring efficient energy use and reducing unnecessary consumption. Under the premise of meeting basic operational requirements, these technologies substantially lower energy consumption per square meter and carbon dioxide emissions in commercial spaces.

From a practical operational perspective, the application of green energy has also driven commercial spaces toward low-carbon and sustainable development. According to research, commercial complexes equipped with high-efficiency lighting systems, smart temperature control technologies, and renewable energy power supplies can reduce carbon emissions by 30% to 50% compared to traditional models. These figures are derived from numerous domestic and international green building certification programs, such as LEED, WELL, and China's Green Building Evaluation Standard, all of which have conducted empirical studies on the carbon reduction effects of green energy in commercial spaces. The promotion of green energy has diversified the energy systems within commercial spaces, thereby enhancing their resilience against energy price fluctuations and extending both economic and environmental benefits over the operational lifecycle.

3.2 Enhancement of Ecosystem Service Functions

The widespread application of green energy technologies plays a vital role in enhancing regional ecological and environmental quality. This is primarily reflected in reducing fossil fuel consumption, lowering pollutant emissions, and improving the functionality of ecosystems such as air, water bodies, and soil. The installation of solar photovoltaic systems and wind turbines effectively reduces carbon dioxide and other greenhouse gas emissions, alleviating the pressure on ecosystems caused by global climate change. Furthermore, the construction of green energy facilities often drives changes in land use practices. Incorporating ecological design elements like vertical greening and rooftop gardens into commercial spaces not only enhances the buildings' environmental performance but also

increases regional biodiversity, thereby providing more stable support for urban ecosystems. Collectively, these measures contribute to the improvement of ecosystem service functions.

4. Practical Approaches and Innovative Models for Spatial Reconstruction

4.1 Ecological Adjustment of Functional Layout

With the advancement of green energy technologies, the functional layout of commercial spaces is shifting from traditional models toward more ecological and efficient approaches. This transformation not only enhances energy utilization efficiency but also improves spatial organization and functional zoning. Technologies such as solar photovoltaic systems, smart lighting control devices, and ground-source heat pumps are employed to reduce energy consumption, improve energy efficiency, decrease reliance on conventional energy sources, and boost overall operational effectiveness. Furthermore, the application of these technologies has led commercial space designs to place greater emphasis on natural lighting, ventilation, and temperature regulation, resulting in functional zones that better adhere to ergonomic principles and enhance user experience within the spaces.

4.2 Integration of User Experience and Environmental Perception

The integration of green energy elements in commercial space design not only enhances spatial sustainability but also profoundly influences users' sensory experiences and behavioral patterns. By employing renewable energy systems such as solar and wind power, along with technologies like smart lighting and energy-efficient air conditioning, commercial environments are improved to foster a more comfortable and healthy living atmosphere. Integrated design subtly incorporates environmental principles into daily life through energy systems, raising public awareness of green concepts and fostering behavioral alignment. Visual representations of green energy—such as real-time energy consumption displays or intuitive visualization of renewable energy usage—deepen users' understanding of spatial ecological value, encouraging eco-conscious choices and enhanced spatial interaction practices.

5. Summary

From the perspective of spatial reconstruction, the integration of green energy guides traditional commercial spaces toward openness, interactivity, and sustainability. Designers

combine energy systems with spatial layouts to create dynamically adaptive environments—featuring adjustable energy management mechanisms, flexible zoning configurations, and multi-dimensional interactive interfaces. This approach not only enhances spatial efficiency but also increases human-space interaction potential. By fully leveraging and strategically utilizing green energy elements, commercial spaces can meet functional requirements while simultaneously improving both ecological value and user experience, thereby establishing a new direction and practical pathway for future commercial space design.

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